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Maximum likelihood network topology identification from edge-based unicast measurements

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Abstract

Network tomography is a process for inferring "internal" link-level delay and loss performance information based on end-to-end (edge) network measurements. These methods require knowledge of the network topology; therefore a first crucial step in the tomography process is topology identification. This paper considers the problem of discovering network topology solely from host-based, unicast measurements, without internal network cooperation. First, we introduce a novel delay-based measurement scheme that does not require clock synchronization, making it more practical than other previous proposals. In contrast to methods that rely on network cooperation , our methodology has the potential to identify layer two elements (provided they are logical topology branching points and induce some measurable delay). Second, we propose a maximum penalized likelihood criterion for topology identification. This is a global optimality criterion, in contrast to other recent proposals for topology identification that employ suboptimal, pair-merging strategies. We develop a novel Markov Chain Monte Carlo (MCMC) procedure for rapid determination of the most likely topologies. The performance of our new probing scheme and identification algorithm is explored through simulation and Internet experiments.

Real-World Impact & Application

Deployable Unicast Internet Topology Identification: Early network tomography required IP multicast, which commercial ISPs blocked for security and billing reasons. Castro et al. made tomography practically deployable across the real Internet by formulating maximum likelihood topology identification using standard unicast packet pairs (sendable from any commodity PC or server).